

Honours Linear Algebra

Week 1 Notes

Jiayu Chen

Systems of Linear Equations, Matrices, Elimination, LU Decomposition, and Vector Spaces

1 Systems of Linear Equations

A linear equation in n unknowns has the form

$$a_1x_1 + \cdots + a_nx_n = b.$$

A system of m such equations can be written as $AX = b$, where A is the coefficient matrix, $X = (x_1, \dots, x_n)^T$, and b is the right-hand side vector.

1.1 A motivating example

Consider

$$\begin{cases} 3x + 2y + z = 39, \\ 2x + 3y + z = 34, \\ x + 2y + 3z = 26. \end{cases}$$

Eliminate the first variable from the second and third equations:

$$E_2 \leftarrow 3E_2 - 2E_1,$$

$$E_3 \leftarrow 3E_3 - E_1.$$

This gives $5y + z = 24$ and $4y + 8z = 39$. Multiplying the second of these by 5 and subtracting 4 times the first gives $36z = 99$. Back substitution then yields

$$z = \frac{11}{4}, \quad y = \frac{17}{4}, \quad x = \frac{17}{4}.$$

2 Matrices and Augmented Matrices

Definition 2.1. An $m \times n$ matrix is a rectangular array with m rows and n columns. The (i, j) -entry is denoted a_{ij} . We write $A = (a_{ij})$.

The coefficient and augmented matrices of a system are respectively

$$A = (a_{ij})_{m \times n}, \quad [A \mid b] = \begin{bmatrix} A & b \end{bmatrix}.$$

Each row of $[A \mid b]$ records one equation. This lets us perform elimination by changing entries rather than repeatedly rewriting variables and equal signs.

2.1 Basic terminology

A $1 \times n$ matrix is a row vector and an $m \times 1$ matrix is a column vector. A square matrix has the same number of rows and columns. The identity matrix I_n has ones on the diagonal and zeros elsewhere. The transpose A^T is obtained by exchanging rows and columns. Upper and lower triangular matrices have zero entries below or above the diagonal, respectively; a diagonal matrix is both.

For matrices of the same size, addition is entrywise:

$$(A + B)_{ij} = a_{ij} + b_{ij}.$$

Scalar multiplication is also entrywise: $(cA)_{ij} = ca_{ij}$. These operations satisfy associativity, commutativity of addition, and the existence of a zero matrix.

3 Elementary Row Operations

The three elementary row operations are:

- (i) interchange two rows;
- (ii) multiply a row by a nonzero scalar;
- (iii) replace a row by itself plus a scalar multiple of another row.

Each operation is reversible. Therefore it preserves the solution set of a linear system. For example, replacing R_i by $R_i + cR_j$ can be undone by replacing R_i by $R_i - cR_j$.

Theorem 3.1 (Equivalence under row operations). *Elementary row operations on an augmented matrix produce an equivalent system: the original and new systems have exactly the same solutions.*

Proof. Row interchange only changes the order of equations. Scaling by a nonzero number replaces an equation by an equivalent one. Replacing one row by itself plus a multiple of another is reversible, so a solution of one system is a solution of the other. $\square$

4 Gaussian Elimination

Gaussian elimination uses elementary row operations to reduce an augmented matrix to a simpler form. A matrix is in row echelon form if:

1. all zero rows are at the bottom;
2. each pivot of a nonzero row is strictly to the right of the pivot above it;
3. every entry below a pivot is zero.

In reduced row echelon form, each pivot is 1 and is the only nonzero entry in its column.

4.1 Complete worked example

For the motivating system, the augmented matrix is

$$\left[\begin{array}{ccc|c} 3 & 2 & 1 & 39 \\ 2 & 3 & 1 & 34 \\ 1 & 2 & 3 & 26 \end{array} \right].$$

The elimination sequence is

$$\left[\begin{array}{ccc|c} 3 & 2 & 1 & 39 \\ 2 & 3 & 1 & 34 \\ 1 & 2 & 3 & 26 \end{array} \right] \xrightarrow{R_2 \leftarrow 3R_2 - 2R_1, R_3 \leftarrow 3R_3 - R_1} \left[\begin{array}{ccc|c} 3 & 2 & 1 & 39 \\ 0 & 5 & 1 & 24 \\ 0 & 4 & 8 & 39 \end{array} \right] \xrightarrow{R_3 \leftarrow 5R_3 - 4R_2} \left[\begin{array}{ccc|c} 3 & 2 & 1 & 39 \\ 0 & 5 & 1 & 24 \\ 0 & 0 & 36 & 99 \end{array} \right].$$

Back substitution gives $z = 11/4$, $y = 17/4$, and $x = 17/4$. The pivots show that every variable is determined, so the solution is unique.

4.2 Reading solutions from echelon form

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

represents $x_1 + 2x_3 = 3$ and $x_2 - x_3 = 4$. The variable x_3 is free. Setting $x_3 = t$ gives

$$X = \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}.$$

By contrast, a row $[0 \ 0 \ 0 \mid c]$ with $c \neq 0$ represents $0 = c$ and proves that the system is inconsistent.

5 Inverse Matrices

Definition 5.1. An $m \times m$ matrix A is invertible if there is a matrix A^{-1} such that $AA^{-1} = A^{-1}A = I_m$.

The inverse, when it exists, is unique. To compute it, row reduce

$$[A \mid I_m] \longrightarrow [I_m \mid A^{-1}].$$

If the left block cannot be reduced to I_m , then A is not invertible.

Proposition 5.2. If A and B are invertible, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

Proof.

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = I,$$

and similarly $(AB)(B^{-1}A^{-1}) = I$. □

Also, $(A^T)^{-1} = (A^{-1})^T$. A triangular matrix with nonzero diagonal entries is invertible; elimination never encounters a zero pivot, and the inverse has the same triangular shape.

6 LU Decomposition

Suppose A can be reduced to an upper triangular matrix U without row interchanges. The elimination matrices satisfy

$$E_k \cdots E_2 E_1 A = U.$$

Therefore

$$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} U = LU,$$

where L is lower triangular with ones on its diagonal.

6.1 Example

For

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{bmatrix},$$

the multipliers are $2/3$, $1/3$, and $4/5$. Hence

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 1/3 & 4/5 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 5/3 & 1/3 \\ 0 & 0 & 12/5 \end{bmatrix},$$

and $A = LU$.

To solve $Ax = b$, first solve $Ld = b$ by forward substitution, then solve $Ux = d$ by back substitution. This is usually preferable to explicitly computing A^{-1} : both approaches have cubic preprocessing cost, but LU is cheaper and generally more stable for repeated right-hand sides.

7 Vector Spaces and Null Spaces

For a homogeneous system $AX = 0$, define

$$\text{Nul}(A) = \{X \in \mathbb{R}^n : AX = 0\}.$$

Theorem 7.1. $\text{Nul}(A)$ is a subspace of $\mathbb{R}^n$.

Proof. The zero vector belongs to $\text{Nul}(A)$. If $u, v \in \text{Nul}(A)$ and $a \in \mathbb{R}$, then

$$A(au + v) = aAu + Av = 0,$$

so $au + v \in \text{Nul}(A)$. Thus the subspace test applies. □

More generally, the span of a set S is the set of all finite linear combinations of vectors in S . It is a subspace, and if U is already a subspace then $\text{Span}(U) = U$.

8 Classifying Solutions

For a system with n variables, the pivot count is at most the number of rows and at most n . A pivot in the augmented column gives a row $[0 \cdots 0 \mid c]$ with $c \neq 0$, which means $0 = c$ and proves inconsistency. Otherwise, non-pivot variables are free.

Theorem 8.1 (Three possibilities). *Every linear system has exactly one of: no solution, one solution, or infinitely many solutions.*

Proof. Row operations preserve the solution set. A contradiction row gives no solution. If there is no contradiction and every variable is a pivot variable, back substitution determines a unique solution. If there is a free variable, assigning different values to it produces infinitely many solutions. $\square$

8.1 A two-parameter family

Suppose reduction gives

$$\left[\begin{array}{cccc|c} 1 & -1 & 4 & 5 & 4 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 3 \end{array} \right].$$

Set $x_3 = s$ and $x_4 = t$. Then $x_1 = 4 - s - 5t$ and $x_2 = 3 + 2s - t$, so

$$X = \begin{bmatrix} 4 \\ 3 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -5 \\ -1 \\ 0 \\ 1 \end{bmatrix}.$$

The two free variables describe two independent directions in the solution set.

8.2 Homogeneous systems

The homogeneous system $AX = 0$ is always consistent because $X = 0$ is a solution. A nonzero solution is called nontrivial.

Theorem 8.2. *An $m \times n$ homogeneous system has a nontrivial solution whenever $n > m$.*

Proof. There are at most m pivots, so at least $n - m > 0$ variables are free. Assigning a nonzero value to one free variable gives a nontrivial solution. $\square$

9 A Second Elimination Example

Solve

$$\begin{cases} x + 2y + 3z = 6, \\ 2x + 3y + z = 7, \\ 3x + 5y + 4z = 13. \end{cases}$$

The augmented matrix becomes

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 2 & 3 & 1 & 7 \\ 3 & 5 & 4 & 13 \end{array} \right] \xrightarrow{R_2-2R_1, R_3-3R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & -1 & -5 & -5 \\ 0 & -1 & -5 & -5 \end{array} \right] \xrightarrow{R_3-R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & -1 & -5 & -5 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

Thus $y + 5z = 5$. With $z = t$, $y = 5 - 5t$ and $x = 5 + 7t$, so

$$X = \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix} + t \begin{bmatrix} 7 \\ -5 \\ 1 \end{bmatrix}.$$

The zero row is not an error: it records a dependent equation.

9.1 An inconsistent system

The reduced matrix

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

contains $0 = 1$, so no choice of variables can solve the system.

10 Matrix Algebra and Geometry

For matrices of the same size, addition and scalar multiplication are entrywise. If $A = [a_1 \cdots a_n]$ and $X = (x_1, \dots, x_n)^T$, then

$$AX = x_1 a_1 + \cdots + x_n a_n.$$

Therefore $AX = b$ is solvable precisely when b lies in the span of the columns of A .

For compatible matrices,

$$(AB)_{ij} = \sum_k a_{ik} b_{kj}.$$

Matrix multiplication is associative but generally not commutative: AB and BA may both exist and be different.

10.1 Elementary matrices

Each elementary row operation is left multiplication by an elementary matrix E . A sequence of operations has the form $E_k \cdots E_1 A$. Since each E_i is invertible, elimination is multiplication by an invertible matrix and does not change the essential solution information.

11 Inverse Matrices in Practice

To compute A^{-1} , row reduce

$$[A \mid I] \longrightarrow [I \mid A^{-1}].$$

For $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$,

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array} \right],$$

so $A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$.

Proposition 11.1. *If A is invertible, then $AX = b$ has the unique solution $X = A^{-1}b$ for every b .*

Proof. Multiplication by A^{-1} gives $X = A^{-1}b$. Conversely, $A(A^{-1}b) = b$. If two solutions existed, their difference would be a nonzero vector in $\text{Nul}(A)$, impossible for an invertible matrix. $\square$

If A and B are invertible, then $(AB)^{-1} = B^{-1}A^{-1}$, and $(A^T)^{-1} = (A^{-1})^T$. A triangular matrix with nonzero diagonal entries is invertible because elimination produces a pivot in every column.

12 LU Decomposition

Without row interchanges, elimination gives $E_k \cdots E_1 A = U$. Therefore

$$A = E_1^{-1} \cdots E_k^{-1} U = LU,$$

where L is lower triangular.

For

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{bmatrix},$$

the elimination multipliers are $2/3$, $1/3$, and $4/5$. Hence

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 1/3 & 4/5 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 3 & 2 & 1 \\ 0 & 5/3 & 1/3 \\ 0 & 0 & 12/5 \end{bmatrix}.$$

Direct multiplication confirms $LU = A$.

To solve $Ax = b$, solve $Ld = b$ by forward substitution, then $Ux = d$ by back substitution. If row exchanges are required, the factorization becomes $PA = LU$.

12.1 A substitution example

For $b = (1, 2, 3)^T$, solve

$$d_1 = 1, \quad \frac{2}{3}d_1 + d_2 = 2, \quad \frac{1}{3}d_1 + \frac{4}{5}d_2 + d_3 = 3.$$

This gives $d = (1, 4/3, 8/3)^T$. The remaining triangular system $Ux = d$ is then solved from the last row upward.

13 Efficiency and Numerical Practice

Both computing an inverse and computing an LU factorization require order n^3 arithmetic operations. Once $A = LU$ is known, each new system with the same A requires only order n^2 operations. Thus LU is especially useful for many right-hand sides.

Explicit inversion can also introduce unnecessary rounding error. Solving $Ld = b$ and $Ux = d$ uses the triangular structure directly and is normally the preferred numerical method. Pivoting avoids unstable divisions by tiny pivots. These observations explain why the theoretical inverse is not always the best computational tool.

14 Null Spaces and Solution Sets

$$\text{Nul}(A) = \{X \in \mathbb{R}^n : AX = 0\}$$

is the null space of A .

Theorem 14.1. $\text{Nul}(A)$ is a subspace of $\mathbb{R}^n$.

Proof. The zero vector belongs to $\text{Nul}(A)$. If $u, v \in \text{Nul}(A)$ and $a \in \mathbb{R}$, then

$$A(au + v) = aAu + Av = 0,$$

so $au + v \in \text{Nul}(A)$. □

If $AX = b$ is consistent and X_0 is one solution, then

$$\{X : AX = b\} = X_0 + \text{Nul}(A).$$

Indeed, $AX = b$ exactly when $A(X - X_0) = 0$. Thus the null space describes all directions in which one may move without changing AX .

15 Span and Subspaces

For $S = \{v_1, \dots, v_r\}$,

$$\text{Span}(S) = \{c_1 v_1 + \dots + c_r v_r : c_i \in \mathbb{R}\}.$$

It is a subspace: the zero combination gives zero, and linear combinations of linear combinations are still linear combinations of the same vectors.

The span of one nonzero vector is a line through the origin. The span of two independent vectors in $\mathbb{R}^3$ is a plane through the origin. The column space of A is the span of its columns, and $AX = b$ is solvable exactly when b belongs to this column space.

15.1 A structural observation

If each u_i is a linear combination of $v_1, \dots, v_r$, then every linear combination of the u_i is also a linear combination of the v_j . This simple observation is the reason that $\text{Span}(S)$ is the smallest subspace containing S .

Problem-solving checklist

1. Form the augmented matrix.
2. Apply reversible row operations.
3. Identify pivots, free variables, and contradiction rows.
4. Parametrize all solutions.
5. For homogeneous systems, interpret the free directions as null-space vectors.
6. For square matrices, use $[A \mid I]$ for inverses or elimination multipliers for LU.

Summary

Systems, matrices, Gaussian elimination, inverses, LU decomposition, null spaces, and spans are connected parts of one theory. Row reduction is the central algorithm: it solves equations, detects

dependence, constructs inverses, produces LU factors, and reveals the geometry of solution sets. The first week develops one central workflow: represent a system by an augmented matrix, apply reversible row operations, read the solution from echelon form, and then interpret the same operations through inverses, LU decomposition, and null spaces. These viewpoints are different descriptions of the same underlying linear structure.